

Kumar Abhishek

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PROFESSIONAL SUMMARY

Generative AI developer with hands-on experience building **RAG systems**, **agentic AI applications**, **MCP servers**, and production-grade **LLM-powered solutions**. Skilled in **LangGraph**, **LangChain**, **FastAPI**, vector databases, and conversational AI systems that integrate external tools and APIs. Experienced in designing end-to-end AI workflows spanning retrieval, reasoning, tool execution, and deployment.

EDUCATION

• Indian Institute of Information Technology, Una <i>B.Tech, Electronics & Communication Engineering</i>	<i>Jul 2023 – Jul 2027</i> CGPA: 7.74/10
• Delhi Model Public School, Delhi <i>Class XII (CBSE)</i>	<i>Jun 2022</i> Score: 85%

PROJECT EXPERIENCE

• InterviewBot – AI-Powered Interview Preparation Platform <i>FastAPI LangChain LangGraph Google Gemini ChromaDB MongoDB</i>	<i>Aug 2024 – Oct 2024</i> Github Live				
- Built a multi-microservice AI interview platform using LangGraph state machines with three distinct conversation modes – Normal Chat (RAG), Interview Setup (multi-turn dialogue), and Contextual Help. - Implemented RAG pipeline with ChromaDB vector store and all-MiniLM-L6-v2 Sentence Transformers for semantic search over website-indexed content; built a resume parser (PDF/DOCX via LangChain) that extracts skills and generates personalized interview questions. - Delivered MCQ test engine and mock interview module with JWT-secured FastAPI backend, async MongoDB (Motor) storage, voice input via React Speech Recognition, and auto-generated PDF performance reports using ReportLab. - Implemented LangSmith tracing and observability to monitor LangGraph workflows, debug execution paths, and evaluate LLM response quality.					
• MoodTunes AI – Agentic Emotion-Based Music Assistant <i>FastAPI React DeepFace Google Gemini DeepGram Last.fm API YouTube Data API</i>	<i>Jul 2024 – Sep 2024</i> Github Live				
- Built an agentic AI music assistant combining conversational AI, voice interaction, emotion recognition, and external tool execution to deliver personalized music experiences. - Integrated Google Gemini with music-search and playback tools, enabling natural-language commands for song search, playlist generation, recommendation workflows, and direct YouTube playback through API-driven tool calls. - Developed a multimodal recommendation pipeline using DeepFace emotion detection, DeepGram speech transcription, Last.fm metadata retrieval, and YouTube playback services to generate context-aware music recommendations.					
• AI Newsletter Automation – FastMCP Framework <i>Python FastMCP arXiv API GitHub API Gmail API Google Drive API Product Hunt API</i>	<i>Oct 2024 – Nov 2024</i> Github				
- Built a production-ready MCP (Model Context Protocol) server using FastMCP exposing 10+ structured tools that enable AI assistants to autonomously perform research, content generation, validation, and newsletter publishing workflows. - Integrated arXiv, GitHub Trending, Product Hunt, Twitter, Gmail, and Google Drive APIs for multi-source content aggregation; implemented OAuth-based Google authentication for Drive auto-save and Gmail feedback analysis across newsletter issues. - Delivered three autonomous workflow phases – Research (fetch papers, repos, products, tweets), Editing (draft creation, section organisation, quality validation), and Publishing (responsive HTML generation, multi-format export) – deployable as a hosted FastMCP Cloud server with zero client-side installation.					
TECHNICAL SKILLS Programming: Python, SQL, Matlab. AI / ML / GenAI: Machine Learning, Deep Learning, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Agentic AI, LangChain, LangGraph, LangSmith, MCP (Model Context Protocol), Transformers, Vector Databases, DeepFace, Agent-to-Agent (A2A) Protocol . Data Analysis & BI: Exploratory Data Analysis (EDA), Statistical Analysis, Hypothesis Testing, Data Cleaning, KPI Development, Dashboard Design, Power BI, Excel (Advanced). Libraries & Frameworks: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PyTorch, Streamlit, SciPy, ReportLab. Backend & Databases: FastAPI, FastMCP, MongoDB, PostgreSQL, ChromaDB Tools & Platforms: GitHub, Jupyter Notebook, Vercel, Hugging Face . Coursework: Machine Learning, Probability & Statistics, Linear Algebra.					
POSITION OF RESPONSIBILITY <table><tr><td>•Active Member, AAVESH Technical Club, IIIT Una</td><td><i>2023 – Present</i></td></tr><tr><td>•Active Member, EUNOIA Literature Club, IIIT Una</td><td><i>2023 – Present</i></td></tr></table>		• Active Member , AAVESH Technical Club, IIIT Una	<i>2023 – Present</i>	• Active Member , EUNOIA Literature Club, IIIT Una	<i>2023 – Present</i>
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